

- 4 (a) (i) Plane polarisation occurs when particles in a wave oscillate only in a single direction perpendicular to the direction of energy transfer of the wave (or direction of wave propagation). [B1]
- (ii) Since sound waves are longitudinal waves, particles in the wave oscillate parallel / along to the direction of energy transfer of the wave (or direction of wave propagation). [B1]

(b) (i) 1. A [B1]

2. $2I$ [B1]

(ii) By Malus' law,

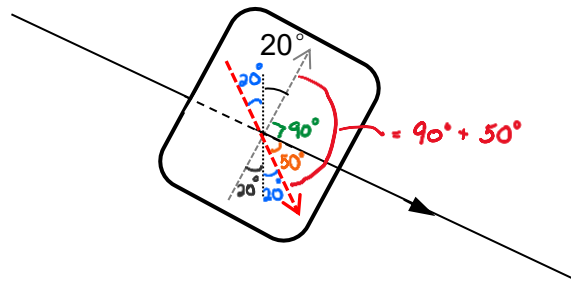
$$I = I_0 \cos^2 \theta$$

$$I_y = I \cos^2 \theta$$

$$= I \cos^2 20^\circ \quad [\text{C1}]$$

$$= 0.883 I \quad [\text{A1}]$$

(iii)



Minimum angle = 140° [B1]

